

Part C

Course title: SC7 Bayes Methods
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1. Let S be a finite set and let $\mathcal{S}$ be the set of all subsets of S . Let $>, <, \sim$ be an order on sets $A, B \in \mathcal{S}$ satisfying $A \sim A$ and $A > B \Rightarrow B < A$. Here are the first three Savage Axioms.

SA1. For any $A, B \in \mathcal{S}$ exactly one of the following relations holds: $A > B$, $A < B$, $A \sim B$.

SA2. For all $A_1, A_2, B_1, B_2 \in \mathcal{S}$, if $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ and $A_i \geq B_i$, $i = 1, 2$ then $A_1 \cup A_2 \geq B_1 \cup B_2$. If in addition either $A_1 > B_1$ or $A_2 > B_2$ then $A_1 \cup A_2 > B_1 \cup B_2$.

SA3. If $A \in \mathcal{S}$ then $\emptyset \leq A$ with $\emptyset < S$.

- (a) [10 marks] Let three sets $A, B, D \in \mathcal{S}$ be given.

- (i) Show that $A \geq B$ implies $A^c \leq B^c$ where A^c is the complement of A in S .

Solution: (B, 3) Suppose $A^c > B^c$. Take $A_1 = A, A_2 = A^c, B_1 = B$ and $B_2 = B^c$. The conditions of SA2 are satisfied so $A \cup A^c > B \cup B^c$ so $S > S$ which is a contradiction. By SA1, A^c and B^c must be ordered so $A^c < B^c$ or $A^c \sim B^c$.

- (ii) Suppose $D \cap A = D \cap B = \emptyset$. Show that $A \geq B$ if and only if $A \cup D \geq B \cup D$.

Solution: (B, 4) Suppose $A \geq B$. Take $A_1 = A, A_2 = D, B_1 = B$ and $B_2 = D$ in SA2. Since $D \leq D$, and the required intersections are empty, the conditions of SA2 are satisfied and hence $A \cup D \geq B \cup D$. Now suppose $A < B$. Take $B_1 = A, B_2 = D, A_1 = B$ and $A_2 = D$ in SA2. Since $A_1 > B_1$ and $A_2 \geq B_2$ the additional condition of SA2 holds so $A \cup D < B \cup D$. Let $X = \{A \geq B\}$ and $Y = \{A \cup D \geq B \cup D\}$. Since $X \Rightarrow Y$ and $\text{not} X \Rightarrow \text{not} Y$, we have $X \Leftrightarrow Y$.

- (iii) Show that $A \subseteq B$ implies $B \geq A$.

Solution: (S, 3) Suppose $A > B$. Take $A_1 = A, A_2 = B \setminus A, B_1 = B$ and $B_2 = \emptyset$. Since $A_1 > B_1$ and $A_2 \geq B_2$ by SA3, we have $A \cup B \setminus A > B \cup \emptyset$, so $B > B$, a contradiction. By SA1, A and B must be ordered so $A < B$ or $A \sim B$.

- (b) [7 marks] Let $z = (z_1, z_2)$ with $z \in S$ and $S = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ be a model index. Consider a linear model with covariates $x_{i,1}, x_{i,2}$, parameters $\beta_1, \beta_2 \in R$, linear predictor $\eta_i = x_{1,i}\beta_1 z_1 + x_{2,i}\beta_2 z_2$ and $y_i \sim p(\cdot | \eta_i)$ jointly independent for $i = 1, \dots, n$ given $\eta_1, \dots, \eta_n$. Let $y = (y_1, \dots, y_n)$.

Assume the marginal likelihoods $p(y|z)$, $z \in S$ are finite and let $\pi(z)$ be the prior and $\pi(z|y)$ the posterior over models $z \in S$. For $A \in \mathcal{S}$ let

$$p(y|A) = \sum_{z \in A} p(y|z)\pi(z|z \in A) \quad \text{and} \quad \pi(A|y) = \sum_{z \in A} \pi(z|y)$$

- (i) Suppose that for all $A, B \in \mathcal{S}$ our preferences satisfy $A < B \Leftrightarrow \pi(A|y) < \pi(B|y)$. Explain briefly why these preferences necessarily satisfy the Savage Axioms.

Solution: (S, 3) Since the Savage Axioms hold if and only if there exists a probability distribution over $(S, \mathcal{S})$ expressing those preference, and we defined our preferences over sets by their order under a probability distribution, the preferences must satisfy the axioms.

- (ii) Suppose now that our preferences over $A, B \in \mathcal{S}$ are given by Bayes factors, so $A < B \Leftrightarrow p(y|A) < p(y|B)$. Under what conditions, if any, do these preferences satisfy the Savage axioms?

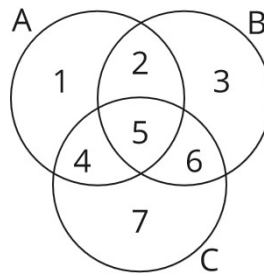
Solution: (N,4) Here $p(y|S) = p(y)$ is the marginal likelihood. Now $A \leq S$ is necessary for all $A \in \mathcal{S}$ by Q1aiii as $A \subseteq S$ so we need $p(y|A) \leq p(y)$ for all $A \in \mathcal{S}$. This implies $\pi(A|y) \leq \pi(A)$ and in particular $\pi(z|y) \leq \pi(z)$ for all $z \in S$. But then $\pi(z|y) = \pi(z)$, $z \in S$, as probabilities are normalised, so $y \perp z$ is necessary (!) for preferences expressed by Bayes factors to satisfy the axioms. But then $p(y|A) = p(y)$ doesn't depend on A . This holds for all $A \neq \emptyset$ as $p(y|\emptyset)$ is undefined. Since we need $A \sim S$ we must have $\emptyset < A$ for all $A \in \mathcal{S}$ by SA3. No probability distribution can express these preferences so they can't satisfy the SAs. For example, for any $A \neq S$ we have $A \sim A^c \sim S$. Take $A_1 = A$, $A_2 = A^c$, $B_1 = S$ and $B_2 = \emptyset$ in SA2. Since $A \sim S$ and $A^c > \emptyset$ we have $A \cup A^c > S \cup \emptyset$ which is a contradiction.

(c) [8 marks] Consider a new axiom SA2':

SA2': for $A, B, C \in \mathcal{S}$ if $A \geq B$ and $B \geq C$ then $A \geq C$.

(i) Show that axioms SA1, SA2 and SA3 imply SA2'.

Solution: (B, 4) The following notation is helpful. Referring to the figure, use the number to represent the corresponding subset. For example $2 = (A \cap B) \setminus C$ and $3 = B \setminus (A \cup C)$. Strings of numbers indicate unions, so $1245 = A$ and $56 = B \cap C$.



The relation $A \geq B$ is $1245 \geq 2356$ and $B \geq C$ is $2356 \geq 5647$. We need to show $1245 \geq 5647$ from SA1, SA2 and SA3. The result in 1aii means we can remove the same set from both sides of an inequality and preserve the inequality. It follows that we can remove the set 25 from both sides of $1245 \geq 2356$ to get $14 \geq 36$ and we can remove 56 from $2356 \geq 5647$ to get $23 \geq 47$. Taking $A_1 = 14$, $A_2 = 23$, $B_1 = 36$ and $B_2 = 47$ and observing $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ we have from SA2 that $1423 \geq 3647$. We can remove 3 then both sides (by 1aii) to get $142 \geq 647$. Since both sides have an empty intersection with 5 so we can add 5 to both sides (by 1aii) and get $1426 \geq 6475$ which is $A \geq C$.

(ii) Show that there exist preferences that satisfy axioms SA1, SA2' and SA3 but do not satisfy SA2 and so SA2 and SA2' are not equivalent.

Hint: take a small set S and specify transitive order relations $>, <, \sim$ satisfying SA1, SA2' and SA3 between all sets in $\mathcal{S}$. Design the relations so they don't satisfy some simple property that follows from SA2.

Solution: (N, 4) If we violate any of the properties from Q1a we must have a contradiction. For example if $S = \{1, 2, 3\}$ and $0 < (1 \sim 3) < 2 < (12 \sim 23) < 13 < 123$ then the relations satisfy SA1, SA2' and SA3. However, if we

take $A = 2$ and $B = 1$ then $A \geq B$ (in fact $>$ so certainly $\geq$) so we must have $B^c \geq A^c$ (as the SA imply this from Q1ai). But $A^c = 13$ and $B^c = 23$ and $23 < 13$ which is a contradiction.

Any set of relations which couldnt possibly be expressed by a probability distribution will do. We saw one above: $\emptyset < A$ and $A \sim S$ for all $A \in \mathcal{S} \setminus \{\emptyset\}$. Those relations are transitive (trivially) and satisfy SA1 and SA3 but we saw in Q1bii that they dont satisfy the SA.

2. (a) [8 marks]

(i) Give the definition of an infinite exchangeable sequence.

Solution: (B,4) Let Z_i , $i \geq 1$ be an infinite sequence of random variables. The sequence is an IES if

$$(Z_1, \dots, Z_n) \sim (Z_{\sigma_1}, \dots, Z_{\sigma_n})$$

for every $n \geq 1$ and every permutation $\sigma_1, \dots, \sigma_n$ of the numbers $1, \dots, n$.

Not required but equivalently...if the Z 's are discrete and $p(z_1, \dots, z_n) = \Pr(Z_1 = z_1, \dots, Z_n = z_n)$ with $z_i \in \mathcal{Z}$ then the displayed equation is

$$p(z_1, \dots, z_n) = p(z_{\sigma_1}, \dots, z_{\sigma_n})$$

for all n and σ and every $(z_1, \dots, z_n) \in \mathcal{Z}^n$. The same relation is sufficient if the Z_i are continuous with joint density $p(z_1, \dots, z_n)$.

(ii) For fixed $K \geq 1$ and $\alpha_j > 0$, $j = 1, \dots, K$ let $\alpha = (\alpha_1, \dots, \alpha_K)$. Let $\Delta_K = \{q \in [0, 1]^K : \sum_{j=1}^K q_j = 1\}$ and $q \sim f(\cdot; \alpha)$ with $f(q; \alpha)$ the Dirichlet density

$$f(q; \alpha) = \frac{\Gamma(\sum_j \alpha_j)}{\prod_j \Gamma(\alpha_j)} \prod_{j=1}^K q_j^{\alpha_j-1}, \quad q \in \Delta_K.$$

For $i \geq 1$ let $Z_i \sim \text{Multinom}(1, q)$ be an infinite sequence of multinomial random variables which are jointly independent given q , so $p(Z_i = j|q) = q_j$ for $j = 1, \dots, K$ and $i \geq 1$. For $z_i \in \{1, \dots, K\}$ let

$$p(z_1, \dots, z_n) = \Pr(Z_1 = z_1, \dots, Z_n = z_n).$$

Let $n_j = \sum_{i=1}^n \mathbb{I}_{z_i=j}$, $j = 1, \dots, K$.

Calculate $p(z_1, \dots, z_n)$ as an explicit function of n, K, α_j and n_j , $j = 1, \dots, K$ and show that Z_i , $i \geq 1$ is an infinite exchangeable sequence.

Solution: (B/S,4 - many very similar examples in notes and PS)

If $f(q; \alpha)$ is the Dirichlet density and $\Delta_K = \{q \in [0, 1]^K : \sum_j q_j = 1\}$ then

$$\begin{aligned} p(z_1, \dots, z_n) &= \int_{\Delta_K} p(z_1, \dots, z_n|q) f(q; \alpha) dq \\ &= \int_{\Delta_K} \frac{\Gamma(\sum_j \alpha_j)}{\prod_j \Gamma(\alpha_j)} \prod_{j=1}^K q_j^{n_j+\alpha_j-1} dq \\ &= \frac{\Gamma(\sum_{j=1}^K \alpha_j) \prod_{j=1}^K \Gamma(\alpha_j + n_j)}{\prod_{j=1}^K \Gamma(\alpha_j) \Gamma(n + \sum_{j=1}^K \alpha_j)}. \end{aligned}$$

Since $p(z_1, \dots, z_n)$ only depends on $z_1, \dots, z_n$ through n_j , the number of times j appears in the first n sampled states, permuting the arguments of $p(z_1, \dots, z_n)$ doesn't change the RHS above, so $p(z_1, \dots, z_n) = p(z_{\sigma_1}, \dots, z_{\sigma_n})$ for any permutation σ of the indices $1, \dots, n$. The random variables exist and their distribution at each n is exchangeable so $Z_1, Z_2, Z_3, \dots$ is an IES.

- (b) [9 marks] Let $x_i \in \{1, \dots, K\}$, $i \geq 1$ be categorical variables with K levels and let $\beta = (\beta_1, \dots, \beta_K)$, $\beta \in R^K$ be a parameter vector. Consider the normal linear model $Y_i = \beta_{x_i} + \epsilon_i$ with $\epsilon_i \sim N(0, \sigma^2)$ jointly independent for $i \geq 1$.
- (i) For $j \in \{1, \dots, K\}$, let $n_j^{(i)} = \sum_{a=1}^i \mathbb{I}_{x_a=j}$ give the number of times level j appears in $x_{1:i} = (x_1, \dots, x_i)$. The x_i -values are a realisation of a sequence of random variables X_i , $i \geq 1$ with $X_1 \sim U\{1, \dots, K\}$ and for $i \geq 1$,

$$\Pr(X_{i+1} = j | X_1 = x_1, \dots, X_i = x_i) = \frac{n_j^{(i)} + 1}{i + K}.$$

Calculate $\pi_X(x_{1:n}) = \Pr(X_1 = x_1, \dots, X_n = x_n)$ and show that $X_1, X_2, X_3, \dots$ is an infinite exchangeable sequence.

Solution: (S/N, 5 - problem sheet problem same but did $K = 2$ so binary, also resembles CRP calculation)

It is possible to do this question by just calculating the product of the conditional probabilities to get the probability for $x_{1:n}$. However there is a certain amount of accounting to do to turn the result into the expression below. We actually did a similar problem this way in a problem sheet and saw that both methods worked.

The X -sequence is just the Z -sequence at $\alpha_j = 1$, $j = 1, \dots, K$. At these α 's,

$$p(z_1, \dots, z_i) = \frac{\Gamma(K)}{\Gamma(i + K)} \prod_{j=1}^K \Gamma(n_j^{(i)} + 1),$$

so that

$$\Pr(Z_{i+1} = j | Z_1 = x_1, \dots, Z_i = x_i) = p(x_1, \dots, x_i, j) / p(x_1, \dots, x_i),$$

(cancel factors using $n_k^{(i)} = n_k^{(i+1)}$ for $k \neq j$)

$$= \frac{\Gamma(i + K)}{\Gamma(i + K + 1)} \frac{\Gamma(n_j^{(i+1)} + 1)}{\Gamma(n_j^{(i)} + 1)}$$

(with $n_j^{(i)} + 1 = n_j^{(i+1)}$ as we have one more copy of j so further cancelation)

$$= (n_j^{(i)} + 1) / (i + K).$$

Since $X_1, X_2, \dots, X_n$ and $Z_1, Z_2, \dots, Z_n$ have the same distribution at every $n \geq 1$, and Z is an IES, X is also.

- (ii) A prior $\pi_B(\beta)$ for $\beta \in R^K$ is given. Let

$$p_{Y|X}(y_1, \dots, y_n | x_1, \dots, x_n) = \int_{R^K} p_{Y|X,\beta}(y_1, \dots, y_n | x_1, \dots, x_n, \beta) \pi_B(\beta) d\beta.$$

Show that, for $\sigma = (\sigma_1, \dots, \sigma_n)$ any permutation of $\{1, 2, \dots, n\}$,

$$p_{Y|X}(y_1, \dots, y_n | x_1, \dots, x_n) = p_{Y|X}(y_{\sigma_1}, \dots, y_{\sigma_n} | x_{\sigma_1}, \dots, x_{\sigma_n})$$

and hence show $(X_1, Y_1), (X_2, Y_2), (X_3, Y_3), \dots$ is an infinite exchangeable sequence.

Solution: (N, 4) For the last part we need to verify

$$((X_1, Y_1), \dots, (X_n, Y_n)) \sim ((X_{\sigma_1}, Y_{\sigma_1}), \dots, (X_{\sigma_n}, Y_{\sigma_n})).$$

The joint distribution is $p(x_1, \dots, x_n)p_{Y|X}(y_1, \dots, y_n|x_1, \dots, x_n)$ so it is sufficient to verify

$$p_{Y|X}(y_{\sigma_1}, \dots, y_{\sigma_n}|x_{\sigma_1}, \dots, x_{\sigma_n}) = p_{Y|X}(y_1, \dots, y_n|x_1, \dots, x_n). \quad (\star)$$

The conditional density of $Y_i|X_i = x_i, \beta$ is $N(y_i; \beta_{x_i}, \sigma^2)$ and the y 's are conditionally independent so

$$p_{Y|X, \beta}(y_1, \dots, y_n|x_1, \dots, x_n, \beta) = \prod_{i=1}^n N(y_i; \beta_{x_i}, \sigma^2)$$

and this is exchangeable for any fixed set of β -values, as permuting the y 's and x 's together just shuffles the factors in the product. The marginal

$$p_{Y|X}(y_1, \dots, y_n|x_1, \dots, x_n) = \int_{R^K} p_{Y|X, \beta}(y_1, \dots, y_n|x_1, \dots, x_n, \beta) \pi_B(\beta) d\beta$$

therefore satisfies $(\star)$ as the conditional under the integral is exchangeable at each $\beta \in R^K$.

(c) [8 marks]

(i) Write down the posterior $\pi(x_{1:n}, \beta|y_{1:n})$ in terms of the model elements.

Solution: (B/S, 3)

$$\pi(x_{1:n}, \beta|y_{1:n}) \propto \pi_B(\beta) \pi_X(x_{1:n}) p_{Y|X, \beta}(y_1, \dots, y_n|x_1, \dots, x_n, \beta)$$

where π_B is given, π_X was calculated in the solution to 2bi and $p_{Y|X, \beta}$ is given in the solution to 2bii.

(ii) Give a Gibbs sampler for $x_1, \dots, x_n$ given $y_{1:n}$ and β .

Solution: (N, 5) The conditional probability $\pi(x_i|x_{-i}, \beta, y_{1:n})$ for $x_i = j$ is

$$\begin{aligned} \pi(x_i = j|x_{-i}, \beta, y_{1:n}) &\propto \pi(x_i = j, x_{-i}, \beta, y_{1:n}) \\ &\propto \pi(x_i = j|x_{-i}) p(y_i|x_i = j, \beta) \\ &\propto (n_j^{(-i)} + 1) N(y_i; \beta_j, \sigma^2) \end{aligned}$$

where $n_j^{(-i)} = \sum_{a \in [n] \setminus \{i\}} \mathbb{I}_{x_a = j}$. Let $\tilde{w}_j^{(i)} = (n_j^{(-i)} + 1) N(y_i; \beta_j, \sigma^2)$. In order to sample this we form the K component vector $w^{(i)} = (w_1^{(i)}, \dots, w_K^{(i)})$ with $w_j^{(i)} = \tilde{w}_j^{(i)} / \sum_k \tilde{w}_k^{(i)}$ and sample $\text{multinom}(1, w^{(i)})$. The Gibbs sampler is

Let $X_t = x_{1:n}$.

(1) Sample $i \sim U\{1, \dots, n\}$ and set $x'_a = x_a$, $a \neq i$.

(2) Sample $x'_i \sim \text{multinom}(1, w^{(i)})$.

(3) Set $X_{t+1} = x'_{1:n}$.

Other versions of the Gibbs sampler such as sequential-scan are fine.

3. (a) [9 marks]

- (i) Let $u \sim g(u)$, $u \in \mathcal{U}$ be a continuous random variable with density g on $\mathcal{U} = R^d$. For $\theta \in \Omega$ with $\Omega = R^p$ and $p \geq d$, let $\theta' = \psi_1(\theta, u)$ where ψ_1 is invertible in u and θ' at fixed θ . As a further condition on ψ_1 , the reverse update $\theta = \psi_1(\theta', u')$ has a unique solution in u' at fixed (θ, θ') . Denote by $u' = \psi_2(\theta, u)$ the solution to $\theta = \psi_1(\psi_1(\theta, u), u')$.

Let $\psi(\theta, u) = (\psi_1(\theta, u), \psi_2(\theta, u))$. Show that $\psi(\psi(\theta, u)) = (\theta, u)$.

Solution: (B,3)

$$\begin{aligned}\psi(\psi(\theta, u)) &= \psi(\psi_1(\theta, u), \psi_2(\theta, u)) \\ &= \psi(\theta', u') \\ &= (\psi_1(\theta', u'), \psi_2(\theta', u')) \\ &= (\theta, \psi_2(\theta', u'))\end{aligned}$$

just using the relations defined in the question statement, so we need to show $u = \psi_2(\theta', u')$. If $\psi_2(\theta', u') = x$ then x solves the relation $\theta' = \psi_1(\psi_1(\theta', u'), x)$ so x is the unique solution to $\theta' = \psi_1(\theta, x)$. But we know $\theta' = \psi_1(\theta, u)$ so $x = u$. It follows that $\psi : \Omega \times \mathcal{U} \rightarrow \Omega \times \mathcal{U}$ is an involution.

- (ii) Assume ψ is differentiable in both arguments. The following MCMC update targets a density $\pi(\theta)$, $\theta \in \Omega$. Let $X_t = \theta$ with $\pi(\theta) > 0$.

- (1) Simulate $u \sim g(u)$ and set $(\theta', u') = \psi(\theta, u)$. Let

$$J_\psi(\theta, u) = \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right|$$

be the Jacobian for the transformation $(\theta', u') = \psi(\theta, u)$.

- (2) With probability

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta, u) \right\}$$

set $X_{t+1} = \theta'$ and otherwise set $X_{t+1} = \theta$.

Show that this update satisfies detailed balance

$$\pi(\theta)g(u)\alpha(\theta'|\theta)d\theta du = \pi(\theta')g(u')\alpha(\theta|\theta')d\theta' du'$$

with respect to $\pi(\theta)$. *Hint: you may assume $J_{\psi^{-1}}(\theta', u') = J_\psi(\theta, u)^{-1}$.*

Solution: (B, 6) Let $\alpha(\theta'|\theta) = \min\{1, r(\theta'|\theta)\}$ and suppose $r \leq 1$. Then $\alpha(\theta|\theta') = \min(1, r(\theta|\theta'))$ where

$$r(\theta|\theta') = \frac{\pi(\theta)g(u)}{\pi(\theta')g(u')} J_\psi(\theta', u')$$

($\psi^{-1} = \psi$ so $J_\psi(\theta', u') = J_{\psi^{-1}}(\theta', u')$ and $J_{\psi^{-1}}(\theta', u') = J_\psi(\theta, u)^{-1}$ by hint)

$$\begin{aligned}&= \left[\frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} \right]^{-1} J_\psi(\theta, u)^{-1} \\ &= r(\theta'|\theta)^{-1}.\end{aligned}$$

It follows that $\alpha(\theta|\theta') = 1$. Starting from the LHS of DB,

$$\begin{aligned}\pi(\theta)g(u)\alpha(\theta'|\theta)d\theta du &= \pi(\theta)g(u)\frac{\pi(\theta')g(u')}{\pi(\theta)g(u)}J_\psi(\theta, u)d\theta du \\ &= \pi(\theta')g(u')\left|\frac{\partial(\theta', u')}{\partial(\theta, u)}\right|d\theta du \\ &= \pi(\theta')g(u')d\theta' du' \\ &= \pi(\theta')g(u')\alpha(\theta|\theta')d\theta' du'\end{aligned}$$

so DB holds (these expressions hold under integrals but the integrals must be equal as the integration domains and integrands are equal).

(b) [8 marks] Let $\pi(\theta|y) \propto \pi(\theta)p(y|\theta)$, $\theta \in R^2$ be a posterior for a given observation model $p(y|\theta)$ and prior $\pi(\theta) = N(\theta; \underline{0}, I_2)$ with $\underline{0} = (0, 0)^T$.

(i) Let $0 < \delta \leq \pi$ (radians) be given. For $\theta \in R^2$ and $u \in [-\delta, \delta]$ let $\theta' = M(u)\theta$ where

$$M(u) = \begin{pmatrix} \cos(u) & -\sin(u) \\ \sin(u) & \cos(u) \end{pmatrix}$$

Here is an MCMC update targeting $\pi(\theta|y)$. Let $X_t = \theta$ with $\pi(\theta|y) > 0$.

1. Simulate $u \sim U(-\delta, \delta)$ and set $\theta' = M(u)\theta$.
2. With probability $\alpha(\theta'|\theta)$ set $X_{t+1} = \theta'$ and otherwise set $X_{t+1} = \theta$.

Calculate the acceptance probability $\alpha(\theta'|\theta)$ simplifying as far as you can.

Solution: (S/N,4) Quoting the result from lectures for the acceptance probability under transformations,

$$\begin{aligned}\alpha(\theta'|\theta) &= \min \left\{ 1, \frac{\pi(\theta'|y)g(u')}{\pi(\theta|y)g(u)} \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right| \right\} \\ &= \min \left\{ 1, \frac{p(y|\theta')}{p(y|\theta)} \right\},\end{aligned}$$

since $g(u') = g(u) = 1/2\delta$ so the g 's cancel. Also, $(M\theta)^T(M\theta) = \theta^T\theta$ in the normal density so $\pi(M\theta) = \pi(\theta)$ and the prior factors cancel. The update is reversed by $u' = -u$ (as $M^T\theta' = \theta$). The Jacobian matrix is therefore

$$\begin{aligned}\frac{\partial(\theta', u')}{\partial(\theta, u)} &= \begin{pmatrix} \partial\theta'/\partial\theta^T & \partial\theta'/\partial u \\ \partial u'/\partial\theta^T & \partial u'/\partial u \end{pmatrix} \\ &= \begin{pmatrix} M & \partial\theta'/\partial u \\ \underline{0}^T & -1 \end{pmatrix}\end{aligned}$$

so the absolute value of determinant of the Jacobian is one (as $\det(M) = 1$ and the zeros in the bottom row save us calculating $\partial\theta'/\partial u$).

(ii) Let $v \sim U(w, 1/w)$ with $0 < w < 1$ and v independent of u . Replace the proposal $\theta' = M(u)\theta$ with $\theta' = vM(u)\theta$. Calculate the new acceptance probability.

Solution: (N/S,4)

$$\begin{aligned}\alpha(\theta'|\theta) &= \min \left\{ 1, \frac{\pi(\theta'|y)g(u',v')}{\pi(\theta|y)g(u,v)} \left| \frac{\partial(\theta',u',v')}{\partial(\theta,u,v)} \right| \right\} \\ &= \min \left\{ 1, e^{-(v^2-1)\theta^T\theta/2} \frac{p(y|\theta')}{p(y|\theta)} \right\},\end{aligned}$$

since $g(u',v') = g(u,v) = 1/2\delta \times 1/(w^{-1} - w)$ so the g 's cancel. Also,

$$(vM\theta)^T(vM\theta) = v^2\theta^T\theta$$

in the normal density so

$$\frac{\pi(\theta')}{\pi(\theta)} = \frac{\exp(-v^2\theta^T\theta/2)}{\exp(-\theta^T\theta/2)},$$

giving the exponential factor involving v^2 .

The inverse is $\theta = v^{-1}M^T\theta'$ so $u' = -u$ as before and $v' = 1/v$. The Jacobian matrix is therefore

$$\begin{aligned}\frac{\partial(\theta',u',v')}{\partial(\theta,u,v)} &= \begin{pmatrix} \partial\theta'/\partial\theta^T & \partial\theta'/\partial u & \partial\theta'/\partial v \\ \partial u'/\partial\theta^T & \partial u'/\partial u & \partial u'/\partial v \\ \partial v'/\partial\theta^T & \partial v'/\partial u & \partial v'/\partial v \end{pmatrix} \\ &= \begin{pmatrix} vM & \partial\theta'/\partial u & M\theta \\ \underline{0}^T & -1 & 0 \\ \underline{0}^T & 0 & -1/v^2 \end{pmatrix}\end{aligned}$$

and $\det(vM) = v^2\det(M) = v^2$ so the absolute value of determinant of the Jacobian is again 1 (product of determinants of diagonal “blocks”).

- (c) [8 marks] Consider two observation models $p_1(y|\phi)$ and $p_2(y|\psi)$ with $\phi \in R^2$ and $\psi \in R^3$ and prior densities $\pi_1(\phi)$ and $\pi_2(\psi)$. The model-prior is $\pi_M(m) = 1/2$, $m \in \{1, 2\}$.
- (i) Give the joint posterior $\pi(\cdot, m|y)$ in terms of the model elements.

Solution: (S,3) The space is $(\theta, m) \in (\{1\} \times R^2) \cup (\{2\} \times R^3)$. The posterior is

$$\pi(\theta, m|y) \propto \begin{cases} \pi_1(\theta)p_1(y|\theta) & \text{when } m = 1 \text{ and } \theta \in R^2, \\ \pi_2(\theta)p_2(y|\theta) & \text{when } m = 2 \text{ and } \theta \in R^3. \end{cases}$$

The normalisation is $2p(y)$ where

$$p(y) = \frac{1}{2} \int_{R^2} \pi_1(\phi)p_1(y|\phi)d\phi + \frac{1}{2} \int_{R^3} \pi_2(\psi)p_2(y|\psi)d\psi.$$

- (ii) Consider the following MCMC update. Let $X_t = (\theta, m)$ with $\pi(\theta, m|y) > 0$.
1. If $m = 1$ simulate $\theta' \sim \pi_2(\cdot)$ and set $m' = 2$, otherwise $\theta' \sim \pi_1(\cdot)$ and $m' = 1$.
 2. With probability

$$\alpha(\theta', m'|\theta, m) = \min \left\{ 1, \frac{p_{m'}(y|\theta')}{p_m(y|\theta)} \right\}$$

set $X_{t+1} = (\theta', m')$ and otherwise set $X_{t+1} = (\theta, m)$.

Write down detailed balance for this update for two states $(\theta, m) = (\phi, 1)$ and $(\theta', m') = (\psi, 2)$ and verify it holds.

Solution: (N,5) In our usual notation in the course, $\rho_{1,2} = \rho_{2,1} = 1$ gives the probability to propose the “other” model. DB requires

$$\pi(d\phi, 1|y)\rho_{1,2}q_{1,2}(d\psi|\phi)\alpha(\psi, 2|\phi, 1) = \pi(d\psi, 2|y)\rho_{2,1}q_{2,1}(d\phi|\psi)\alpha(\phi, 1|\psi, 2)$$

to hold “under the integral”. Here the proposals are $q_{1,2}(d\psi|\phi) = \pi_2(\psi)d\psi$ and $q_{2,1}(d\phi|\psi) = \pi_1(\phi)d\phi$. We should verify that

$$\int_{\phi \in A} \int_{\psi \in B} \pi(\phi, 1|y)\pi_2(\psi)\alpha(\psi, 2|\phi, 1)d\psi d\phi = \int_{\psi \in B} \int_{\phi \in A} \pi(\psi, 2|y)\pi_1(\phi)\alpha(\phi, 1|\psi, 2)d\phi d\psi$$

holds for all sets $A \subset R^2$ and $B \subset R^3$ for which the integrals are defined.

Because the integration domains are the same on both sides, the measures are the same, and we can exchange the integrals as the integrals exist, it is sufficient that the integrands match at every $(\phi, \psi) \in R^2 \times R^3$, that is

$$\pi(\phi, 1|y)\pi_2(\psi)\alpha(\psi, 2|\phi, 1) = \pi(\psi, 2|y)\pi_1(\phi)\alpha(\phi, 1|\psi, 2).$$

Substituting for $\alpha(\theta', m'|\theta, m)$ on the LHS of DB assuming $p_2(y|\psi)/p_1(y|\phi) \leq 1$,

$$\begin{aligned} \pi(\phi, 1|y)\pi_2(\psi)\alpha(\psi, 2|\phi, 1) &= \frac{\pi_1(\phi)p_1(y|\phi)}{2p(y)}\pi_2(\psi)\frac{p_2(y|\psi)}{p_1(y|\phi)} \\ &= \pi(\psi, 2|y)\pi_1(\phi)\alpha(\phi, 1|\psi, 2) \end{aligned}$$

since $\alpha(\phi, 1|\psi, 2) = 1$. The proof for $p_2(y|\psi)/p_1(y|\phi) \geq 1$ is the same but with $(\phi, 1) \leftrightarrow (\psi, 2)$.